PW6 — 2D median nerve model

Selected values, why they are used, and where they enter the MATLAB model. The cross-section is a synthetic first-pass geometry, not a patient-specific reconstruction.

1. Anatomy and electrode layout

Choice / code step	Value	Reason	Source
Model level	Mid-upper arm, before forearm motor branches	Places the stimulation section proximal to the forearm branches for the selected forearm-motor target.	Delgado-Martínez et al. (2016), median-nerve fascicular topography and branching.
Rnerve	HCS area 9.20 ± 1.73 mm²; equivalent-circle radius 1.71 mm	Radius is calculated as √(A/π) from the mid-upper-arm histology mean. The circle is a 2D simplification.	Pušnik et al. (2025), histological cross-sections (HCS).
fasc	10 fascicles; total area 5.01 mm²; largest 1.49 mm²	Uses the HCS mean count and total area. The layout is one large and nine equal fascicles to fit the section; it is synthetic, not a traced specimen.	Pušnik et al. (2025), HCS.
zCut	−10.3 mm	Selects one supplied longitudinal device station. This is a device coordinate, not a distance from the elbow.	Professor's `sites.mat` and `load_active_sites.m`.
Active contacts	Gold; 50 × 200 × 200 µm	Uses the slide specification. The code overrides the 5 µm thickness in `sites.mat` with 50 µm; positions and orientations remain from the supplied data.	Professor's course slides and supplied site file.
Cuff contacts	On nerve surface; no squeeze	Contact centre is set to `Rnerve + thickness/2`. The nerve is not deliberately reduced. The cuff sleeve and mechanical pressure are not represented in this 2D model.	Group design choice; site data from professor.

In MATLAB, the site data are loaded first, the contacts at zCut are selected, and their 2D outlines are added to the PDE geometry. Their boundaries are retained for later electrode boundary conditions. Gold is treated as an ideal contact boundary; no bulk gold conductivity is assigned as tissue.

2. Tissue conductivities

The 2D section uses sigmaT (x–y plane). sigmaL is kept for a future 3D model; it does not enter this transverse 2D solve.

Tissue (code order)	σT [S/m]	σL [S/m]	Reason and source
Endoneurium	0.0826	0.571	Adopted anisotropic peripheral-nerve FEM values. Zelechowski et al. (2020); parameter context in Pelot et al. (2019).
Perineurium	0.00088	0.00088	Highly resistive sheath; close to Pelot et al.'s DC model estimate and matches the adopted FEM baseline in Zelechowski et al.
Epineurium	0.0826	0.0826	Value assigned in Zelechowski et al.'s FEM; a modeling assumption, not a direct human median-nerve measurement.
Surrounding skeletal muscle	0.13	0.56	Assumes muscle fibres parallel to nerve. The 2D plane uses conductivity perpendicular to fibres; the longitudinal value is retained for 3D. Kangasmaa et al. (2025).

Mesh refinement: the perineurium is 14 µm thick, so hPeri = 0.005 mm gives several elements across the sheath. Electrode edges use hElec = 0.01 mm. These are numerical mesh settings, not tissue measurements.

References

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Delgado-Martínez I, et al. "Fascicular Topography of the Human Median Nerve for Neuroprosthetic Surgery." *Frontiers in Neuroscience*, 2016;10:286. DOI: 10.3389/fnins.2016.00286.

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